Solving Vehicle Routing Problem Using Grover Adaptive Search Algorithm

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***Abstract*— Intelligent Transportation Systems (ITS) occupy a pivotal position in the contemporary societal landscape, with one of their widespread applications being the vehicle routing problem, renowned for its classification as an NP-hard challenge. Traditional algorithms often require considerable time to address such problems or can only provide approximate solutions. Recent progress in quantum computation, however, offer a promising alternative by utilizing the parallelism inherent in quantum mechanics. In this study, we propose to solve the vehicle routing problem using the Grover adaptive algorithm. Compared to classical algorithms and existing quantum methods, our innovative algorithm achieves a quadratic speedup, with a time complexity of $O(K\sqrt{\frac{N}{M}})$. Additionally, it is applicable to a broader range of real-world scenarios, as it can directly handle linear inequality constraints without introducing slack variables. Notably, the algorithm operates independently of classical optimizers and gradient-based methods, thereby circumventing potential challenges such as the "barren plateaus" phenomenon. Experimental results from four instances demonstrate the feasibility and correctness of the proposed approach, highlighting its capability to handle inequality constraints easily.**

***Index Terms*— Quantum computation, vehicle routing problem, Grover adaptive algorithm.**

I. Introduction

NOWADAYS, characterized by population explosion and an expanding range of transportation choices, traffic congestion has become a progressively severe issue. In this context, intelligent transportation systems (ITS) [1], [2], [3], [4], [5] play a vital role. The primary goal of ITS is to achieve safe, efficient, and energy-efficient traffic planning and scheduling. This could be achieved by seamlessly integrating advanced methods in communications and computation into the transportation system. Over the past few decades, ITS has been widely implemented in navigation, railway, and air transport systems, serving as a fundamental component of the modern intelligent society [6], [7].

Traffic scheduling is inherently a complex combinatorial optimization problem (COP) [8], [9]. These problems exhibit an exponential increase in the number of feasible solutions with the scale of the problem, making them challenging to solve in polynomial time. For instance, the vehicle routing problem (VRP), a commonly encountered application of ITS, is classified as an NP-hard problem [10], [11]. The existing algorithms for VRP can generally be categorized into three groups: (1) Accurate algorithms. Common accurate algorithms, such as dynamic programming [12], [13] and branch-and-bound [14], solve problems using precise mathematical methods. While these methods theoretically guarantee finding the optimal solution and often outperform artificial intelligence algorithms due to their comprehensive and systematic search approach, they require substantial computational effort as the problem size increases, resulting in exponentially growing resource demands. As a result, these algorithms are typically only feasible for small to medium-sized VRP instances. Additionally, they are often tailored to specific problems, limiting their general applicability. (2) Heuristic algorithms [15], [16]. Common heuristic algorithms, such as the greedy algorithm [17], solve problems using heuristic information and experiential rules, rather than strict mathematical models. These algorithms are relatively simple and easy to implement and provide acceptable solutions, making them suitable for solving large problems. However, they often perform local searches, focusing on the current best solution, which can lead to relatively poor-quality solutions as they may get trapped in local optima. (3) Meta-heuristic algorithms [18], [19]. Common meta-heuristic algorithms, such as simulated annealing [20], extend heuristic approaches by incorporating randomness, enabling global searches and avoiding local optima. This usually results in higher solution quality compared to heuristic methods. However, meta-heuristic algorithms require numerous iterations, leading to longer computation times, and their performance is highly sensitive to parameter settings. Each method has its advantages, but they often become computationally expensive or fail to provide satisfactory solutions within a reasonable time as the problem size grows.

Quantum computation [21], [22], [23], [24], [25] offers promising solutions to address this challenging scenario, facilitated by its parallel computing capabilities rooted in

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physical phenomena such as quantum superposition and entanglement [26]. Several approaches have been proposed for solving the Vehicle Routing Problem (VRP) using algorithms such as quantum annealing (QA) [27], [28], quantum approximate optimization algorithm (QAOA) [29], and variational quantum eigensolver(VQE) [30], [31]. The QA [32], [33] algorithm is an optimization algorithm based on quantum tunneling effects, which evolves the initial quantum state into the final state corresponding to the optimization problem's solution. In practical applications, the algorithm's performance is influenced by a series of tunable control parameters, such as problem size, to achieve optimal result. However, the adjustments may be challenging as each control parameter may be closely interdependent. Despite efforts to alleviate this issue [34], [35], [36], the additional computational time and resources required for parameter tuning remain inevitable. The QAOA [37] is a hybrid classical-quantum algorithm particularly suited for noisy intermediate-scale quantum (NISQ) devices and is frequently used for solving combinatorial optimization problems, providing approximate solutions. The fundamental principle of QAOA involves alternating the application of the problem Hamiltonian, which encodes the objective function and constraints, and the mixing Hamiltonian on an initial state. A classical optimizer is then used to adjust parameters to minimize the objective function. The performance and training time of QAOA are primarily influenced by the hyperparameters such as the number of circuit layers, the type of the optimizer used, and the quality of initial parameters [29]. Therefore, considerable time must be invested in carefully tuning these hyperparameters to optimize the algorithm's overall performance [38], [39]. while increasing the number of layers can improve the model's performance, the reliance on gradient-based optimization introduces the risk of vanishing gradients, which can hinder training and prevent the algorithm from finding a solution. The VQE is another quantum-classical hybrid algorithm adapted for optimization problems. The key distinction between VQE and QAOA lies in the circuit layout: QAOA employs a particular ansatz, whereas VQE allows for arbitrary ansatzs. This flexibility enables VQE to potentially select more efficient ansatzes for optimization problems, often resulting in better performance [40]. However, both VQE and QAOA are variational quantum algorithms (VQAs) [41] and thus face the challenge of barren plateaus [42], where the optimization landscape becomes flat, making it difficult to find the optimal solution.

The Grover search algorithm (GSA) [43] is specifically designed to search through an unstructured database of N entries, achieving a query complexity of $O(\sqrt{\frac{N}{M}})$, where M represents the number of marked target elements. This represents a significant improvement compared to the classical approach, which on average requires $\frac{N}{2M}$ queries. GSA provides a polynomial speedup over classical algorithms, making it a crucial subroutine in various applications, including quantum machine learning (QML) [44] and combinatorial optimization problems, where it ensures a guaranteed speedup. For instance, building upon the Grover algorithm, a novel adaptive global optimization algorithm, known as Grover adaptive search (GAS) [45], [46], [47], hlhas been developed to identify the minimum value in a database.

In this paper, we present an exploration that uses GAS to solve VRP. First, we assign an initial threshold T to the objective function f of the VRP, which is treated as the current minimum dispatch cost. Next, the VRP is transformed into the quadratic unconstrained binary optimization (QUBO) form. Using a method known as the quantum dictionary (QD) [48], we encode the "new objective function" of the VRP in its QUBO form onto the quantum state, thereby completing the preparation of the initial database state. This new objective function is obtained by subtracting the threshold from the original objective function. In the subsequent step, the constraints of the VRP are encoded within the $oracle$ operator, which marks the target states. The phase-tagging operation in the $oracle$ is executed only when all constraints are satisfied and $f - T$ is negative, amplifying the amplitudes of the corresponding states that represent feasible solutions with lower dispatch costs. This process is repeated, with each newly identified lower cost serving as the updated threshold, until the optimal solution is found or a predefined stopping condition is met. Given that the number of solutions M is unknown in each iteration, we employ the randomized technique described in [49], ensuring that solutions are found in $O(\sqrt{\frac{N}{M}})$ time. Finally, we evaluate our proposed scheme through rigorous testing on various instances.

Compared to the above quantum vehicle routing approaches, the key advantages of our work are as follows:

- First, the time complexity of our algorithm is $O(K\sqrt{\frac{N}{M}})$, where K represents the number of iterations and N represents the total number of possible combinations of binary decision variables in the QUBO formulation of the VRP. This results a quadratic speedup compared to other classical or quantum algorithms designed to solve the VRP in the same problem formulation. Furthermore, this complexity is mainly influenced by the choice of the first initial threshold $T_{initial}$. By reasonably estimating the range of the objective function values and selecting a smaller initial threshold, the number of iterations K can be significantly reduced. The time complexity of this algorithm, if analogous to the metrics used in wireless routing systems, could be mapped to routing overhead—typically referring to the total cost of the path used to transmit data from the source network to the destination network.
- Second, our algorithm requires no hyperparameters that demand extensive tuning time, except for the first initial threshold $T_{initial}$. In comparison to algorithms such as QA, our approach significantly reduces the overall execution time. Moreover, $T_{initial}$ can be selected randomly without requiring prior estimation. This choice would only moderately increase the number of iterations but it does not affect the quality of the algorithm's solution.
- Third, our algorithm for solving VRP does not involve any classical optimizers or gradient calculations. This effectively avoids issues related to gradients, such as training stagnation caused by barren plateaus.

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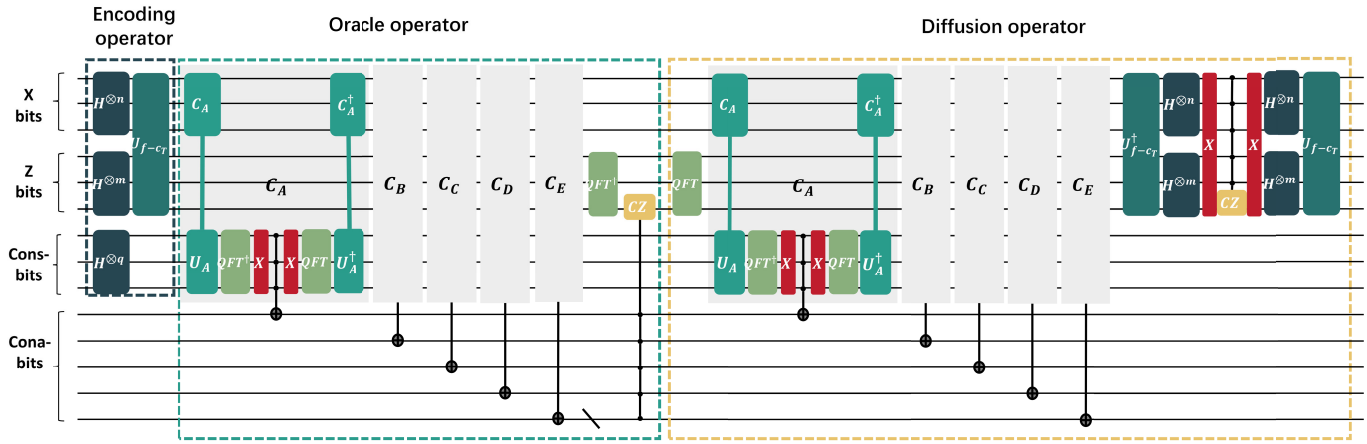


Fig. 1. The detailed diagram of the scheme. Distinct qubits are allocated for encoding variables(X bits), the objective function(Z bits), constraints($Cons$ bits), and auxiliary bits($Cona$ bits), with their quantities determined by the parameters set for the VRP. In the $Oracle$ operator, unitary operator C_A, C_B, C_C, C_D, and C_E encode the constraints in Eq.(12)-(16). To conserve quantum computation resources, different constraints reuse the $Cons$ bits. For instance, after inverting one $Cona$ bit, C_A resets the $Cons$ bits. In the $diffusion$ operator, C_A, C_B, C_C, C_D, and C_E are applied once again to reset $Cona$ bits for reuse in subsequent processes. Before measurement, a $QFT^{\dagger}$ operation is performed on the Z bits.

- Fourth, the existing QA, QAOA, and VQE encode constraints by converting them into quadratic penalty terms [50]. Consequently, they are only capable of handling linear equality constraints and cannot directly manage inequality constraints, such as $\sum_{i=1}^{n_x} A_i x_i \leq b$. When dealing with inequality constraints, these algorithms must introduce enough additional binary decision variables, known as slack variables, to convert the inequality constraints into equality constraints. For example, for $\sum_{i=1}^{n_x} A_i x_i \leq b$, the number of slack variables required is $n_\lambda = \lceil 1 + \log 2\ b \rceil$, and the converted constraints becomes $(\sum_{i=1}^{n_x} A_i x_i - b + \sum_{j=1}^{n_\lambda} 2^j \lambda_j)^2$. In contrast, our proposed method can directly address both linear equality and inequality constraints without the need for additional decision variables or the conversion of inequality constraints into quadratic penalty terms. This capability is particularly significant as inequality constraints are common in practical applications of the VRP. For example, in variants of the VRP that more accurately reflect real-world scenarios, such as the capacitated vehicle routing problem (CVRP), the vehicle's capacity must be below a specified limit.

Furthermore, we compare our work with a protocol designed based on the Grover algorithm [51]. The fundamental idea of this protocol is to utilize the Grover algorithm to select the optimal next-hop node at each node until the destination in the network is reached. The comparison centers on the practicality and implementation difficulty of the GSA in routing optimization problems and yields the following results:

- Firstly, in the protocol, the number of target solutions is known to be one for each search execution, whereas in our work, the number of solutions is unknown. This underscores the practicality of our approach, as in most real-world routing optimization problems, the solutions are unknown.
- Secondly, the construction of the $Oracle$ operator in the protocol is implicit, requiring knowledge of the search answer to complete the phase-tagging operation within the $Oracle$ operator. This clearly prevents the GSA from the practical use. In contrast, as illustrated in Fig. 1, the $Oracle$ operator in this paper is explicitly constructed using the inversion-test technique [52], without the need for prior knowledge of the answer.

II. Theoretical Framework

In this section, we initially explore the models of VRP and GAS algorithms, serving as the foundation for our quantum vehicle routing scheme. Subsequently, we conduct an in-depth analysis of VRP before introducing our proposed solution.

A. The Issue of Vehicle Routing

The VRP is a fundamental topic in combinatorial optimization, with significant implications for both economic advancement and academic inquiry. The objective is to determine the optimal route for a fleet of identical trucks to serve a set of dispersed customers located around a central depot. Each customer is visited exactly once, and all trucks must start and finish their routes at the depot. The optimal route is the one that satisfies all constraints while minimizing the total length of the routes. Mathematically, the VRP with n locations (including the depot) and k vehicles is formally denoted as $(n,\ k)$.

B. The Grover Search Algorithm

The Grover search algorithm is specifically designed to locate a specific target item in an unsorted quantum database. The detailed steps of the Grover search algorithm are outlined as follows:

- Initializing the quantum state
  A quantum register comprising n qubits is initialized, where n denotes the number of qubits needed to encode the items in the database.

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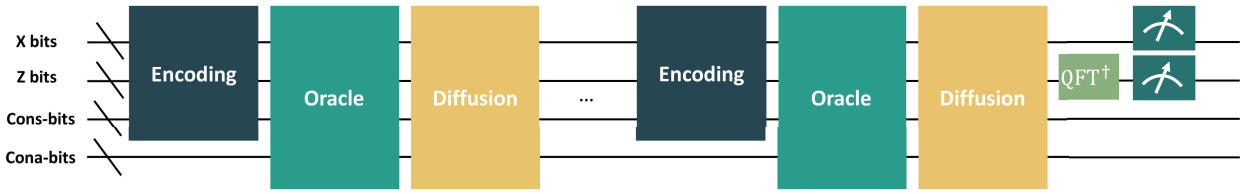


Fig. 2. The schematic diagram of our scheme.

- Loading the classical database onto the quantum state
  A classical database of size N is loaded onto an n-qubit uniform superposition state. The unitary operator responsible for this loading, denoted as U_{data}, usually consists of n Hadamard gates applied to the $|0\rangle^{\otimes n}$ state. The procedure of this operation is described as follows:

$$U_{data}\,|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle\,. \tag{1}$$

- The searching process
  a. Apply the $Oracle$ operator O to mark the target item. The $Oracle$ is a one-way function defined as:

$$O\frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} |i\rangle = -\,|target\rangle + |nontarget\rangle\,, \tag{2}$$

  where $|target\rangle$ denotes the target state and $|nontarget\rangle$ represents the set of non-target states.
  b. Apply the $diffusion$ operator to invert the phase of the $|0\rangle^{\otimes n}$ state. This step, commonly referred to as 'amplitude amplification', involves rotating the phase of the $|0\rangle^{n}$ state by 180°, thereby amplifying the probability amplitude of the target state while compressing the amplitude of the non-target states. The combination of these two processes is known as the Grover search and can be symbolized as the G operator.
- Iterating the Grover search process t times
  The Grover search operation is repeated t times to maximize the probability amplitude of the target state. The number of iterations, t, depends on the proportion λ of the target state in the database. If λ is unknown, quantum counting algorithms [53] or methods mentioned in Refs. [54] can be applied.
- The readout process
  Finally, measure and observe the state of the quantum register. The probability amplitude of the target item is amplified, and the measurement outcome will reveal the target item.

C. The Grover Adaptive Search Algorithm

The Grover adaptive search algorithm (GAS) is a stochastic algorithm based on the Grover search algorithm, suitable for finite optimization problems. In each iteration, the algorithm employs Grover search technology to identify data where the objective function value if greater than (or less than) the current known maximum (or minimum). This iteration process continues until the optimal value is found or a predefined termination condition is met. Specifically, the working principle of GAS involves treating the set of all possible objective function values as a database, where all data exceeding (or falling short of) the current threshold are treated as target data. Therefore, the algorithm can amplify the probability amplitude of the target data through the $Oracle$ and $diffusion$ operators, ultimately facilitating the discovery of the optimal value.

D. Solving VRP Using the Grover Adaptive Search Algorithm

In this part, we present a comprehensive framework for addressing the VRP using the GAS algorithm. First, we outline the essential components of the VRP, including the objective function and constraints that define the problem. The VRP can be classified into its basic version and various extensions that incorporate additional constraints. This study primarily focuses on the fundamental VRP, with supplementary discussions on extended versions provided in a later section. Next, we reformulate the VRP as a constrained polynomial binary optimization (CPBO) model. Finally, we detail the procedures for solving the VRP using the GAS algorithm.

Initially, we provide a detailed introduction to the problem:

- The objective function of the VRP is typically defined as:

$$f_{Obj} = \sum_{i=0,\,j=0,\,i\neq j}^{n-1} w_{ij}x_{ij}. \tag{3}$$

  Here w_{ij} denotes the distance between two locations(customers), and the unit transportation cost is set to 1. x_{ij} is a binary variable indicating the presence of a vehicle route between the two locations. For example, if the vehicle moves to customer 2 after visiting customer 1, then x_{12} is assigned a value of 1; otherwise, it is set to 0. This means that x_{ij} represents an edge between locations i and j, where i is the starting point and j is the destination. All binary variables collectively form the vector $\mathbf{x}$ of size $n(n-1)$, expressed as:

$$\mathbf{x} = (x_{01}, x_{02}, \cdots, x_{0n-1}, x_{10}, x_{12}, \cdots, x_{n-1n-2}). \tag{4}$$

- Here, we outline the constraints of the problem:
  (1) To guarantee that all vehicles both depart from and return to the depot at location 0, it is imperative that the number of paths originating from the depot equals k, as well as the number of paths terminating at the depot. These constraints can be effectively represented by the following two equations:

$$\sum_{j\in S[0]} x_{0j} = k, \tag{5}$$

$$\sum_{j\in E[0]} x_{j0} = k. \tag{6}$$

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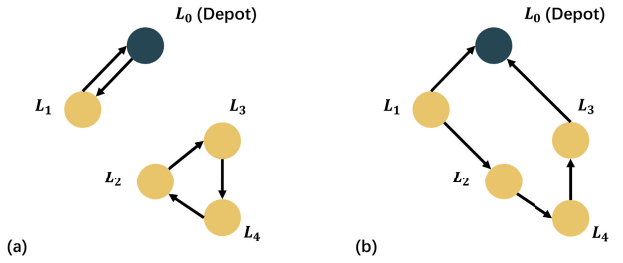


Fig. 3. The left diagram represents a scheduling route that includes a sub-loop, while the right diagram illustrates an actual feasible scheduling route.

Here, the set $S[0]$ contains the locations j to which node 0 sends an edge, while the set $E[0]$ comprises the locations j which sends an edge to 0.

(2) To ensure that each customer is visited exactly once, for each location, there must be exactly one path starting from that location and one path ending at that location. The corresponding mathematical expression is:

$$\sum_{j\in S[i]} x_{ij} = 1 \qquad \forall i \in \{1,\cdots,n-1\}; \tag{7}$$

$$\sum_{j\in E[i]} x_{ji} = 1 \qquad \forall i \in \{1,\cdots,n-1\}. \tag{8}$$

(3) Furthermore, to prevent the formation of sub-loops within the scheduling routes, as depicted in Fig. 3, the following constraints are necessary:

$$\sum_{i,j\in I} x_{ij} \le |I|-1, \qquad 2 \le |I| \le n-1, I \subset L. \tag{9}$$

Here, L denotes the set of all locations. This equation ensures that only cycles incorporating all locations, i.e., $|I| = n$ cycles, are retained, while all other cycles are eliminated. For convenience, we denote the aforementioned four constraints as C_A, C_B, C_C, C_D, and C_E, respectively.

- CPBO formulation of the VRP

The CPBO is an extension of the well-known quadratic unconstrained binary optimization (QUBO) problem. The QUBO problem is utilized in a diverse range of disciplines, including finance and artificial intelligence. Particularly, traditional problems in computer science, such as maximum cut and graph coloring, can be reformulated as QUBO problems. The primary objective is to determine the optimal assignment of binary values to a collection of n_v variables, represented as $\mathbf{x}$, to minimize the objective function. The general form of the objective function can be expressed as follows:

$$\min_{x\in\{0,1\}} \left(\mathbf{x}Q^\top\mathbf{x} + g^\top\mathbf{x} + c\right), \tag{10}$$

where Q is an $n_v \times n_v$ two-dimensional matrix, g is an n_v-dimensional vector and c is a constant term. There are a total of $2^{n(n-1)}$ possible distributions of vector $\mathbf{x}$ in Eq.(4).

Next, we present the QUBO formulation of the objective function f_{Obj} and the corresponding constraint problems. By combining these two formulas, we can derive the CPBO formulation for VRP as:

$$\min_{x\in\{0,1\}} \left(f_{Obj}^{CPBO} = \mathbf{W}\mathbf{x}\right) \tag{11}$$

$$s.t. \mathbf{x}Q_{C_A}^\top\mathbf{x} + g_{C_A}^\top\mathbf{x} + c_{C_A} = 0, \tag{12}$$

$$\mathbf{x}Q_{C_B}^\top\mathbf{x} + g_{C_B}^\top\mathbf{x} + c_{C_B} = 0, \tag{13}$$

$$\mathbf{x}Q_{C_C}^\top\mathbf{x} + g_{C_C}^\top\mathbf{x} + c_{C_C} = 0, \tag{14}$$

$$\mathbf{x}Q_{C_D}^\top\mathbf{x} + g_{C_D}^\top\mathbf{x} + c_{C_D} = 0, \tag{15}$$

$$\mathbf{x}Q_{C_E}^\top\mathbf{x} + g_{C_E}^\top\mathbf{x} + c_{C_E} \le 0, \tag{16}$$

in which, Eq. (12)-(16) are not directly derived from Eq. (5)-(9), but rather from the equivalent transformation of equations (5)-(9), i.e., Eq. (17)-(21).

$$\left(\sum_{j\in S[0]} x_{0j} - k\right)^2 = 0, \tag{17}$$

$$\left(\sum_{j\in E[0]} x_{j0} - k\right)^2 = 0, \tag{18}$$

$$\left(\sum_{j\in S[i]} x_{ij} - 1\right)^2 = 0 \ \forall i \in \{1,\cdots,n-1\}, \tag{19}$$

$$\left(\sum_{j\in E[i]} x_{ji} - 1\right)^2 = 0 \ \forall i \in \{1,\cdots,n-1\}, \tag{20}$$

$$\left[\sum_{i,j\in I} x_{ij} - (|I|-1)\right]^2 \le 0, 2 \le |I| \le n-1, I \subset L. \tag{21}$$

Then, we elaborate the process of our methodology (as shown in Fig. 1). This method determines the value set of variables that minimize the objective function in the VRP through the following three steps:

1) State preparation and threshold initialization

The first step in performing classical tasks with quantum algorithms is to load classical data onto quantum states. Basis encoding involves converting the given classical data into binary form and then mapping each binary digit to a single qubit. For example, the classical data $'8'$ can be encoded into the quantum state $|111\rangle$ using basis encoding.

Furthermore, to load each set of binary variable values and their corresponding objective function values onto quantum states in a one-to-one correspondence, we adopt a method known as the quantum dictionary. This method entangles two quantum registers to represent the quantum state as a dictionary of key-value pairs: one register for keys and the other for values, as described in [48] and [55]. In this work, we take the binary variables x_{ij} as keys, the corresponding objective function values as values, accomplishing state preparation using this method. For clarity, we define the symbols as follows: the qubits in the key register are denoted as X bits, those in the value register as Z bits, and the qubits used for encoding constraints as $Cons$ bits and $Cona$ bits, respectively.

Based on the above description, the encoding process for the objective function f_{Obj}^{CPBO} can be summarized as follows. First, use X qubit as control qubits to apply a

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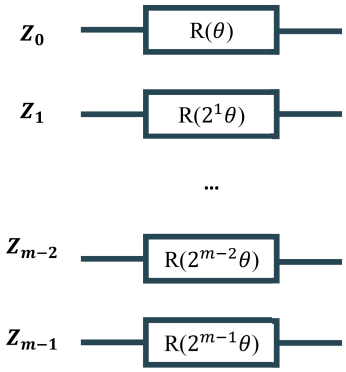


Fig. 4. The encoding circuit $QC(\theta)$ for an integer. $R(\theta)$ denotes the phase gate. m is the number of qubits used, chosen based on the magnitude of the integer.

sequence of controlled phase gates to the Z qubits, with the phase gates taking effect only when the target qubit is in the state $|1\rangle$. Next, apply the inverse quantum Fourier transform (QFT) to the Z qubits. After completing the encoding, the value register is in a uniform superposition state representing all possible objective function values. To enhance the clarity of the encoding process, consider the following example of constant term encoding. Given an integer $-2^{m_Z} \leq c \leq 2^{m_Z}$, where m_Z represents the size of the Z bits, the integer c can be encoded onto the state $|c \mod 2^{m_Z}\rangle_Z$ using the quantum circuit $QC(\theta)$ as illustrated in Fig. 4, where $\theta \in [-\pi, \pi)$. In this circuit, θ is set to $\frac{2\pi}{2^{m_Z}}c$, and the encoding is completed with the inverse QFT.

When applied to a uniform superposition state, the effect of $QC(\theta)$ is as follows:

$$QC(\theta)\frac{1}{\sqrt{2^{m_Z}}}\sum_{i=0}^{2^{m_Z}-1}|i\rangle_Z = \frac{1}{\sqrt{2^{m_Z}}}\sum_{i=0}^{2^{m_Z}-1}e^{ic\theta}|c\rangle_Z . \tag{22}$$

When encoding integers using this method, the result is the binary two's complement of the integer [55]. For negative integers, the highest bit of the Z bits will be $|1\rangle$; otherwise, it will be $|0\rangle$. For example, with $m_Z = 3$, the integer -1 would be encoded as $|111\rangle_{m_Z}$. Consequently, we refer to the highest bit of both Z bits and $Cons$ bits as the 'flag' qubit. This flag qubit indicates the sign of the current encoded values of Z bits and $Cons$ bits.

The coefficients of quadratic and linear terms can also be encoded using the method described above. For instance, a quadratic term $a_{12}V_1V_2$, involving two binary variables, V_1 and V_2, can be encoded using the circuit shown in Fig. 5.

In this work, to solve the VRP, we encode the objective function as $f_{Obj}^{CPBO} - c_T$, where c_T is a self-defined scheduling cost threshold. Initially, we estimate the range of possible scheduling costs and select a value randomly within this range to serve as the threshold c_T. At this point, we have completed the preparation of the objective function state $f_{Obj}^{CPBO} - c_T$, and the corresponding circuit is shown in Fig. 1.

2) Grover search

In this step, the Grover search operation is employed to find a scheduling route that satisfies the constraints and minimizes the scheduling cost.

Traditionally, the Grover search operation consists of the $oracle$ and $diffusion$ operator. Here the $oracle$ operator is composed of two components: the first part encodes the constraints onto the $Cons$ bits, while the second part assesses the satisfaction of these constraints and the sign of $f_{Obj}^{CPBO} - c_T$. If both conditions are met, a controlled-phase gate is applied to phase-tag the corresponding group of binary variables. The encoding in the first part is accomplished using the method previously described, namely the quantum dictionary. In the second part, constraints assessment is performed according to Eq.(12)-(15), where $Cons$ bits should be in the state $|0\rangle_{Cons}^{\otimes m_{Cons}}$ (m_{Cons} represents the size of the $Cons$ bits), and for Eq.(16), the flag bit of $Cons$ bits should be $|1\rangle$. In our design, if the constraints in Eq.(12)-(16) are satisfied, then $Cona$ bits will undergo the following transformation:

$$|0\rangle_{Cona}^{m_{Cona}} \rightarrow |1\rangle_{Cona}^{m_{Cona}}, \tag{23}$$

where m_{Cona} denotes the size of the $Cona$ bits. Furthermore, if $f_{Obj}^{CPBO} - c_T$ is negative (i.e.,$f_{Obj}^{CPBO} - c_T$<0), it implies the existence of a target function value smaller than the current threshold; in this case, the flag qubit of the Z bits will be $|1\rangle$.

In summary, when all conditions are met, $Cona$ bits and the flag qubit of the Z bits will be $|1\rangle_{Cona}^{m_{Cona}}$ and $|1\rangle$, repectively. Based on this, we perform the multi-controlled Z gate to complete the phase-tagging. The construction method of the $diffusion$ operator is illustrated in Fig. 1. Combining $diffusion$ operator and $Oracle$ operator amplifies the amplitudes of states corresponding to routes that satisfy all conditions. To maximize the amplitudes of these states, the Grover search must be repeated r times. Since the proportion of solutions, λ, is unknown, r is determined using the method described in Refs. [49].

3) Measurement and threshold update

In this step, we perform measurements on two sets of quantum registers: X bits and Z bits. Based on these measurements and the previous threshold, we calculate the current minimum objective function and set it as the new threshold for subsequent iterations of the algorithm. If none of the amplitudes corresponding to any state are amplified, it indicates that no scheduling routes have costs below the current threshold. In this case, the threshold can be randomly reinitialized, or we can check whether the predetermined termination conditions of the algorithm are met. The above steps are then repeated until the termination conditions are satisfied.

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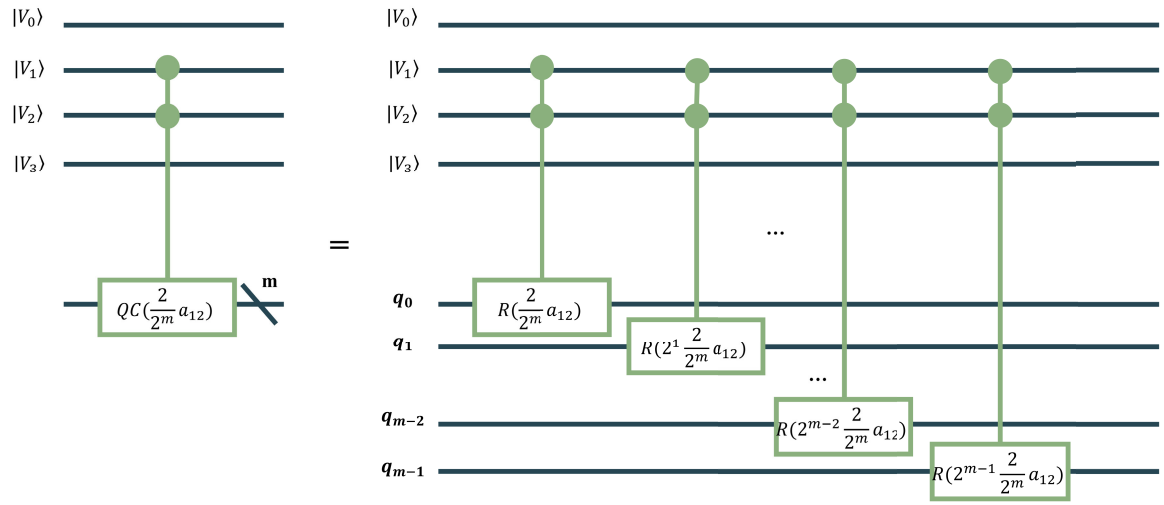


Fig. 5. The encoding circuit for the term $'a_{12}V_1V_2'$. The encoding methods for quadratic and linear terms used in this paper are similar to this.

III. The Performance of the Scheme

We analyze the performance of our algorithm using several VRP examples of varying scales as case studies. First, we present the problem sizes and parameter settings for these instances. Next, we describe the experimental framework employed during the simulation process. Finally, we illustrate the results with visual representations and provide an in-depth analysis. To ensure the reproducibility and reliability of our results, all experiments were conducted in the following environment. All experiments were performed on a Windows 10 operating system. The hardware configuration included a $12th$ Gen Intel(R) Core(TM) i9-12900K@3.20 GHz processor, 32.0 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of VRAM. Additionally, the experiments were executed on the pyQPanda platform, a quantum computation cloud platform developed by Origin Quantum, with Python version 3.7.

- (3, 1): The VRP with three locations and one vehicle
  According to Eq.(4), the number of required binary variables is $O(poly(n))$, where n represents the number of locations. Thus, both this instance and the next instance involve 6 binary variables. Given that we use basis encoding for these variables, the number of X bits is $m_X = 6$. Considering the limited computational capabilities of existing quantum computing cloud platforms, we have randomly generated weights (rounded for simplicity) for this and next instance: $W_{(3,k)}$: $(69, 59, 69, 21, 59, 21)$, as illustrated in Fig. 6(a). Due to the relatively smaller number of X bits, the number of Z bits can be bigger, for example, $m_Z = 9$. Note that the constraints encoded in Fig. 1 correspond to those outlined in Eq.(12)-(16). For problems involving 3 locations, the constraints in Eq.(12)-(15) are sufficient to eliminate the possibility of sub-loop issues. Therefore, only the constraints in Eq.(12)-(15) need to be encoded, resulting in $m_{Cona} = 4$ for the $Cona$ bits. From the form of the constraints in Eq.(17)-(20), it is evident that the elements of the coefficients matrix encoded in Eq.(12)-(15) will not exceed 4. Consequently, the number of $Cons$ bits is $m_{Cons} = 3$.
  For a VRP with size $(3, k)$, where $k = 1$ or 2, there is only one feasible solution that satisfies the conditions. Consequently, this instance and the next one primarily serve to validate the feasibility of applying the GAS algorithm to solve the VRP. In each iteration of the algorithm, the Grover search operation is executed 5 times. The results are depicted in Fig. 7. Since the total number of measurement outcomes is 2^6, Fig. 7 only present the top 20 measurement outcomes with probabilities greater than 10^{-4}.
  To demonstrate the algorithm's capability, we deliberately set a small initial threshold value of 100. This threshold is lower than the unique feasible solution that we have, aiming to test the algorithm's capability to accurately identify scheduling costs below the current threshold. The experimental results, shown in Fig. 7, confirm our expectations: as no output is smaller than the threshold, no probability amplitudes of any states are amplified.
  Subsequently, we reinitialize the threshold to a larger value of 250, and the algorithm successfully identifies the unique feasible solution. The probabilities corresponding to this solution are significantly amplified, while those of other infeasible solutions are suppressed. Using this unique feasible solution as the new threshold, the algorithm continues its search. The results indicate that there are no new feasible solutions below this threshold, leading to the termination of the algorithm. Overall, the algorithm demonstrates its ability to determine the existence of feasible solutions and locate them with high probability. The choice of the initial threshold and its subsequent refinement based on the identified solutions are crucial to the algorithm's success.
- (3, 2): VRP with three locations and two vehicles
  Except for the difference in the number of vehicles, all other parameter settings for this instance remain

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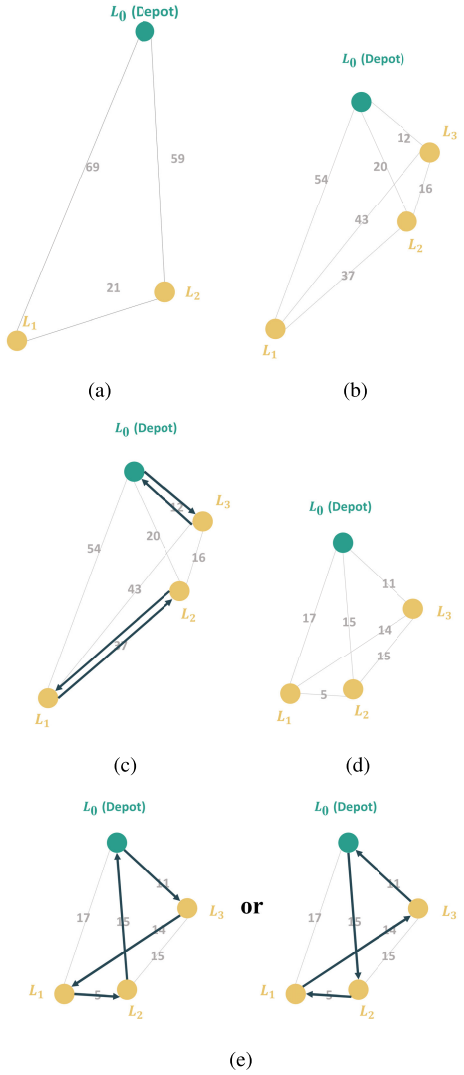


Fig. 6. (a) Graph for (3, 1) and (3, 2) VRP. (b) An example used to showcase the effects of the algorithm without imposing constraint conditions in Eq.(24). (c) The optimal schedule routes found for the aforementioned example, involving sub-loop. (d) Graph for (4, 1) VRP. (e) The optimal schedule routes found for the (4, 1) VRP. The gray numbers in the figures represent the distances between different locations.

consistent with those used in the previous model. The results are shown in Fig. 8.

From the preliminary results, it is obvious that the algorithm solve the VRP correctly while satisfying the required constraints. It is noteworthy that at this scale, sub-loop issues do not arise, and only equality constraints are involved. The treatment of inequality constraints will be discussed in the next instance.

- (4, 1): VRP with four locations and one vehicle

The number of binary variables involved in this instance, as well as the next one, is 12; hence, the parameter is set to be $m_X = 12$. This instance introduce the potential issue of sub-loops, requiring the $Oracle$ to encode the conditions in Eq.(16). Specifically, the following set of equations is sufficient to ensure the absence of sub-loops:

$$\begin{aligned}(x_{12}+x_{21})^2-2<0,\\(x_{13}+x_{31})^2-2<0,\\(x_{23}+x_{32})^2-2<0.\end{aligned}\tag{24}$$

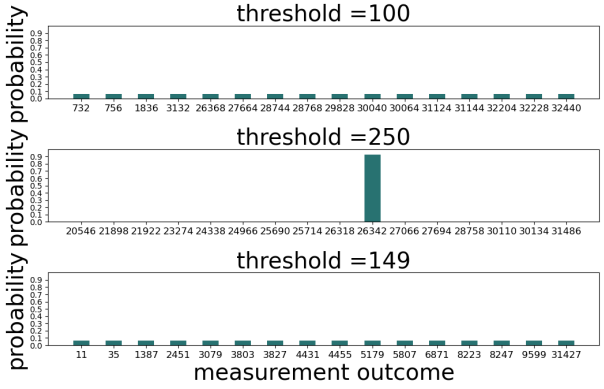


Fig. 7. The iterations process for (3, 1) VRP. The vertical axis represents measurement probabilities, and the horizontal axis indicates measurement outcomes in decimal form. Expand the horizontal axis into binary form, the last 6 positions represent the values of binary variables, i.e., the scheduling route, while the remaining qubits represent the binary complement of the $f-c_T$ for that route. The process reveals that the optimal scheduling solution for this problem is $L_0 \rightarrow L_2 \rightarrow L_1 \rightarrow L_0$, with a scheduling cost of 149.

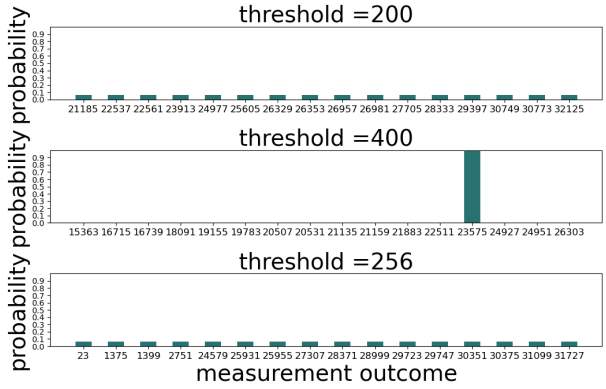


Fig. 8. The iterations process for (3, 2) VRP. From the process shown, the optimal scheduling solution is determined to be $L_0 \rightarrow L_1 \rightarrow L_0, L_0 \rightarrow L_2 \rightarrow L_0$, with a scheduling cost of 256.

If the above three conditions are not taken into account, the optimal scheduling route found by the algorithm for the larger weights, as depicted in Fig. 6(b), would be the one shown in Fig. 6(c). This route satisfies the constraints in Eq.(12)-(15), but it fails to eliminate sub-loops, rendering it impractical for real-life scenarios. Consequently, it is necessary to include the conditions outlined in Eq.(24), resulting in a required number of $m_{Cona} = 7$ bits. The number of $Cons$ bits remains $m_{Cons} = 3$. Additionally, the number of Z bits is set to $m_Z = 7$, and the generated weight vector is $W_{(4,1)} = (17, 15, 11, 17, 5, 14, 15, 5, 15, 11, 14, 15)$, as shown in Fig. 6(d). The iterative process of the algorithm is illustrated in Fig. 9, which presents only the top 20 measurement outcomes with probabilities greater than 10^{-5}. The optimal scheduling route identified by the algorithm is shown in Fig. 6, with the minimum scheduling cost being 45.

To further highlight the effectiveness of Eq.(24) in eliminating sub-loops, we examine an example with smaller weights, as shown in Fig. 10, presenting the optimal route solution and the corresponding iterative process in Fig. 10 and Fig. 11, respectively. The results show that the

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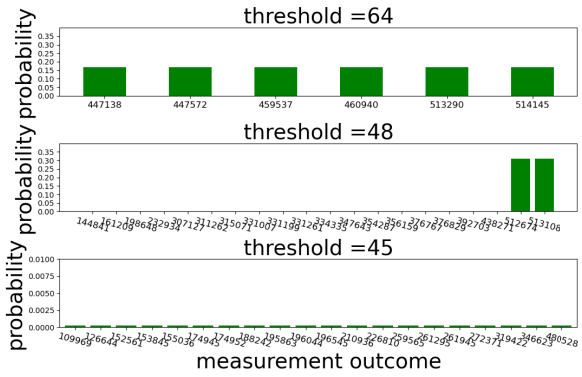


Fig. 9. The iterations process for (4, 1) VRP. The vertical axis represents measurement probabilities, while the horizontal axis indicates measurement outcomes in decimal form. Expand the horizontal axis into binary form, the last 12 positions represent the values of binary variables, i.e., the scheduling route, while the remaining qubits represent the binary complement of the $f - c_T$ for that route. Based on this process, the optimal scheduling solution for this problem, as shown in Fig. 6, has a scheduling cost of 45.

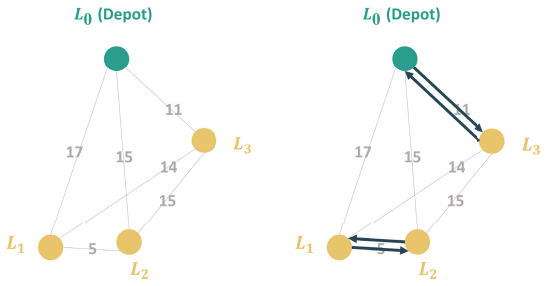


Fig. 10. The optimal schedule routes found for the (4, 1) VRP without applying the constraints in Eq.(24).

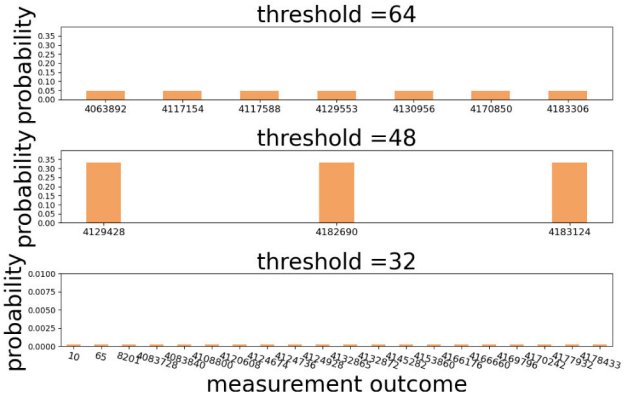


Fig. 11. The iterations process for the (4, 1) VRP without the constraints in Eq.(24). From this process, the optimal scheduling solution for this problem, as shown in Fig. 10, has a scheduling cost of 32.

optimal route found by the algorithm includes sub-loops, thus making it an impractical solution.

- (4, 2): VRP with four locations and two vehicles
  Since this instance does not involve sub-loops, we set $m_{Cona} = 4$. The number of X bits and $Cons$ bits remains the same as in the case of (4, 1). Therefore, the parameter m_Z is set to 10, and the weight $W_{(4,2)}$ is defined as:(50, 12, 69, 50, 40, 20, 12, 40, 59, 69, 20, 59). The iterative process of the algorithm is illustrated in Fig. 12, and further details are not reiterated.

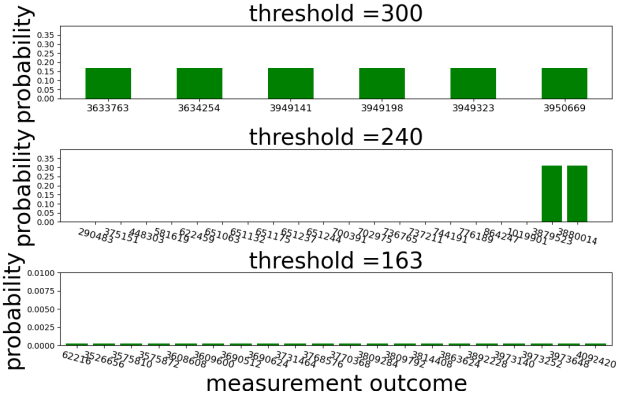


Fig. 12. The iterations process for (4, 2) VRP. From this process, we can determine the optimal scheduling solution and the corresponding scheduling cost for this problem.

The four instances presented above clearly demonstrate the effectiveness of our method in tackling the fundamental version of the VRP. However, when dealing with more complex variants of the VRP, such as the Capacitated VRP (CVRP), additional considerations must be incorporated. In the CVRP, each vehicle has a limited capacity to meet the demands of multiple customers, and the available locations are restricted accordingly. To adapt our approach for the CVRP, we can simply expand our method. Specifically, quantum registers and the $Oracle$ operator will be extended to capture and process the additional information. This extension allows the algorithm to make informed decisions during the routing process by considering the specific demands and available locations of each customer. Thus, our proposed approach serves as a solid foundation for addressing a wide range of VRP variants, adapting to their complexity and specific requirements.

IV. Conclusion

In summary, we propose a novel approach to solving the VRP by employing the GAS algorithm. The key elements of this work involve using the $Oracle$ operator to identify feasible solutions with costs below the current threshold. These solutions are then amplified in their probabilities, resulting in a new threshold. Compared to existing quantum vehicle routing algorithms, our approach offers several advantages. Firstly, our scheme provides a quadratic speedup, with a time complexity of $O(K\sqrt{\frac{N}{M}})$. Secondly, the algorithm does not require extensive time for parameter tuning, as the iterative process involves only one hyperparameter, $T_{initial}$. Even if $T_{initial}$ is chosen randomly, it does not affect the quality of the final solution. Moreover, since our method does not involve any classical optimizers or gradient calculations, it avoids issues such as vanishing gradients. Additionally, our approach can directly handle linear inequality constraints without introducing slack variables to convert inequalities into equalities. This makes our approach more efficient in terms of binary decision variable bits and more applicable in real-world scenarios with complex inequality constraints. Finally, our algorithm is capable of finding the global optimal solution rather than an approximate one. The corresponding code is available at https://github.com/

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lutonggui/lutonggui-Solving-Vehicle-Routing-Problem-Using-Grover-adaptive-search-algorithm. While our research contributes valuable insights into the domain of quantum computation for solving classical problems, it is important to acknowledge its limitations. For instance, our algorithm is currently applicable to solving small-scale VRP, and it may pose a challenge for existing noisy intermediate-scale quantum(NISQ) [56] devices when dealing with larger problem instances.

In the future, exploring more efficient encoding methods could potentially overcome these challenges and advance our understanding of solving combinatorial optimization problems based on the algorithm. We believe this study contributes to the ongoing dialogue on accelerating classical problem-solving with quantum computation and lays the groundwork for further research in related domains.

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